AdifySkills

n8n Automation Mastery
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CLASS NOTES

🔌 Module 3 · APIs, Webhooks & Integrations

DAY 10 of 25

Databases: Postgres, Supabase, Airtable & Notion

"Sheets se aage — real databases ke saath scale karo."

⚡ AAJ HUM YEH BANAYENGE

Supabase Leads API (Create / Update / Search)

Ek hi webhook se teen kaam: action = create, update ya search. Data Supabase (Postgres) table mein jaata hai aur API JSON lautati hai — apna chhota backend bina server ke.

Postgres · Supabase · Airtable · Notion

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

Sheets 10,000 rows ke baad slow ho jaati hain. Real clients ko real database chahiye — Postgres/Supabase.

🎯 Aaj ke baad aap yeh kar paoge

01 SQL basics: SELECT, INSERT, UPDATE, JOIN

🌍 Librarian se: "Delhi wale saare students do" (SELECT), "naya student jodo" (INSERT).
⚙️ SELECT name, phone FROM leads WHERE city = 'Delhi';

02 Postgres/MySQL nodes & Supabase

🌍 Bank ka main computer system.
⚙️ Postgres node → Execute Query / Insert / Upsert; Supabase node → rows CRUD.

03 Airtable & Notion as no-code databases

🌍 Sundar register (Airtable), digital diary (Notion).
⚙️ Airtable → Create record; Notion → Database page create.

04 Upsert patterns & deduplication

🌍 Contact list mein same number do baar save na ho.
⚙️ ON CONFLICT (email) DO UPDATE … / Remove Duplicates node.

05 n8n Data Tables (built-in storage)

🌍 Ghar mein chhota locker — bahar jaane ki zarurat nahi.
⚙️ n8n Data Tables: chhote projects ka built-in storage.

06 Choosing the right DB for a client

🌍 Chhote kaam ke liye cycle, bade ke liye truck.
⚙️ < 5k rows + non-tech → Airtable; app backend / scale → Postgres/Supabase.

⏱ Class ka plan (2 ghante)

10m	35m	45m	20m	10m

`0:00-0:10` **Recap**

🪄 *Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.*

`0:10-0:45` **Concept: Databases**

Sheets vs Airtable vs Postgres

🪄 *Notebook mein 10 naam dhoondhna easy, 10,000 mein mushkil. Database ek super-fast librarian hai.*

`0:45-1:30` **Live Build**

Mini CRM on Supabase with n8n API layer

🪄 *Aaj ki kahani yaad hai — 'Library aur smart librarian'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.*

`1:30-1:50` **Quiz + Challenge**

Module 3 quiz

🪄 *Practice match — jitna yahan girenge, utna asli match mein kam girenge.*

`1:50-2:00` **Q&A + Homework**

🪄 *Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?*

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

Table, row, column

Database = Excel jaisa par fast aur reliable.

🪄 **Analogy:** Library ki almirah aur shelves

CONCEPT 02

SQL basics

SELECT * FROM leads WHERE city = 'Delhi'

🪄 **Analogy:** Librarian se sawaal poochna

CONCEPT 03

Upsert

Unique key (email) match → update, warna insert.

🪄 **Analogy:** Attendance: naam hai to tick, nahi to naya naam

CONCEPT 04

No-code DBs

Airtable/Notion — non-tech clients ke liye easy UI.

🪄 **Analogy:** Ready-made furniture

🔑 Aaj ka main idea

Jab data bada ho jaata hai (hazaaron rows), Sheets slow aur risky ho jaati hain. Database (Postgres/Supabase) fast, safe aur multi-user hai. Airtable/Notion beech ka raasta hain — non-tech client ke liye easy UI.

- Sheets = notebook, Database = bank ka ledger system
- Airtable = sundar register jismein filters
- Upsert = attendance: naam hai to tick, nahi to add

📖 Ek kahani se samjho: Library aur smart librarian

1 📚 **Library = database** = Postgres / Supabase

2 🗂️ **Almirah = table** = leads table

3 📖 **Kitaab = row** = Ek lead record

4 🙋 **Librarian se sawaal** = SQL SELECT

5 🔁 **Kitaab hai to update, nahi to nayi** = Upsert

🪄 Ek line mein: “Notebook mein 10 naam dhoondhna easy, 10,000 mein mushkil. Database ek super-fast librarian hai.”

🗺️ Aaj ka workflow: Supabase Leads API (Create / Update / Search)

Data left se right chalta hai. Har dabba ek n8n node hai.



💰 Isse kitna kama sakte ho?

Startups (mini backend / leads API) — aam taur par **₹15,000 – ₹40,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1. ⚡ **Webhook**
   POST /crm/lead
2. 🗄 **Supabase**
   Upsert into leads (email unique)
3. ⚡ **Webhook**
   GET /crm/search?city=
4. 🗄 **Postgres**
   SELECT name, phone FROM leads WHERE city = $1
5. ⚡ **Respond to Webhook**
   Results JSON

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- ☐ Connect: **Supabase (project URL + service key)**
- ☐ Supabase SQL editor mein yeh chalao: create table leads (id bigint generated always as identity primary key, name text, email text unique, status text default 'new', created_at timestamptz default now());

✅ Test kaise karein

1. curl se action=create bhejo, phir search, phir update
2. Supabase Table Editor mein data dekho

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😫 Pehle (manual)

10,000 rows ki slow Sheet	10 min
Duplicate dhoondhna	20 min
	30 min

⚡ Ab (n8n)

DB mein fast search + auto dedupe

0 min

⏱ Search seconds mein, duplicates automatic band.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo 📋 Select karke copy karo

```
INSERT INTO leads (email, name, city) VALUES ($1, $2, $3)
ON CONFLICT (email) DO UPDATE SET name = EXCLUDED.name, city = EXCLUDED.city;
```

📦 Sample data — input se output tak

📥 Input 📋 Select karke copy karo

```
{
  "email": "raj@mail.com",
  "name": "Raj",
  "city": "Jaipur"
}
```

📤 Output 📋 Select karke copy karo

```
[
  {
    "name": "Raj",
    "phone": "9000011111",
    "city": "Jaipur"
  }
]
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Ek hi webhook se teen kaam: action = create, update ya search. Data Supabase (Postgres) table mein jaata hai aur API JSON lautati hai — apna chhota backend bina server ke.

Flow: Webhook (POST /leads-api) → Switch (action?) → Supabase (create / update / search) → Respond (JSON result)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 Debug helper prompt 📋 Select karke copy karo

```
I'm building the n8n workflow "Supabase Leads API (Create / Update / Search)" (nodes: POST /leads-api, Which Action?, Create Lead, Update Status, Search Leads, Respond).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

Workflow JSON
6 nodes · ready to import
`adifyskills.com/workflows/day-10-supabase-leads-api.json`

Lesson page
Copy buttons, quiz aur video links
`adifyskills.com/day/10`

Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Supabase (project URL + service key)" connect karo.
Webhook/Form test par kuch nahi ho raha?	"Listen for test event" / Test URL use karo aur 2 minute ke andar request bhejo. Production URL sirf Active workflow par chalta hai.

⚠️ Yeh galtiyan sabse common hain

- ! SQL mein user input seedha jodna (SQL injection) — query parameters use karo
- ! Unique constraint ke bina upsert
- ! Supabase service key ko frontend mein expose karna

⭐ Pro tips — time bachane wali tricks

- ★ n8n Data Tables chhote projects ke liye built-in storage
- ★ Index lagao frequently searched columns par
- ★ Client ke liye Airtable interface bana do — woh khud data dekh sake

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Supabase mein leads table + n8n se create/update/search endpoints.

📝 Homework

Notion content calendar ko Airtable ke saath 2-way sync karo.

- ☐ Homework ho gaya

BEGINNER
Airtable mein record add

INTERMEDIATE
Notion ↔ Airtable sync

ADVANCED
Paginated search API with LIMIT/OFFSET

🧩 Aaj ka challenge — pehle khud try karo

🎯 Same email se dobara "create" aaye to error ki jagah "already exists" message lautao.

💡 **Hint 1**
Pehle search karo, phir decide karo create karna hai ya nahi.

💡 **Hint 2**
Create se pehle Supabase getAll (filter email=eq.X) + IF "item mila?"

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Data padhne ka SQL command?

(A) DROP
(B) UPDATE
(C) INSERT
(D) SELECT

Q2. SQL injection se bachne ke liye?

(A) Capital letters
(B) Long query
(C) Comments
(D) Query parameters

Q3. Non-tech client ke liye easy DB?

(A) Postgres CLI
(B) Airtable
(C) Redis
(D) MongoDB shell

Q4. Upsert ka matlab?

(A) Backup
(B) Delete+insert
(C) Update ya insert
(D) Sirf insert

📖 Aaj ke naye shabd

Webhook
Ek URL jis par koi doosra app data bhejta hai aur aapka workflow turant chal padta hai.

JSON
Data likhne ka format: { "naam": "Ravi", "city": "Delhi" } — key aur value ke jode.

CRM
Customers aur leads ka record rakhne ka system.

✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 11

Lead Generation & CRM Automation

"Har lead, har baar, 60 seconds ke andar follow-up."

☐ **Day 10 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → D
SELECT

Q2 → D
Query parameters

Q3 → B
Airtable

Q4 → C
Update ya insert

✅ Challenge ka solution

Which Action? (create) → Supabase getAll (email=eq.{{ $json.body.email }}, Always Output Data on) → IF {{ $json.id }} exists → Respond "already exists", warna Create Lead.

📚 Official docs

- Postgres node ↗
- Supabase node ↗
- Notion node ↗
- Airtable node ↗

▶ Videos

- n8n supabase tutorial ↗
- n8n postgres node ↗

💬 Doubt poochho
WhatsApp: +91 83838 91716
wa.me/918383891716 ↗

🌐 Day 10 lesson page
Quiz, copy buttons, videos
adifyskills.com/day/10 ↗

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